
Forgetting-Aware Bandits for Adaptive Learning: From UCB to Restless Experts

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Abstract

Adaptive question selection in educational systems is fundamentally limited by *forgetting*: knowledge decays over time for topics that are not reinforced, rendering static selection policies suboptimal. We present a comprehensive study of 16 bandit algorithms for adaptive question selection under Ebbinghaus forgetting dynamics, framing the problem as a restless multi-armed bandit (RMAB). We introduce four novel algorithms—Forgetting-UCB (F-UCB), which augments exploration bonuses with decay-aware urgency; BKT-Bandit, which maintains Bayesian posteriors with integrated forgetting; Advantage Index, which computes budget-aware indices via MDP value differences; and MetaSelector, a UCB-over-experts approach that interpolates across regimes online. We extend RMAB-based approaches to a comprehensive quizzing benchmark with Ebbinghaus forgetting dynamics and identify the *Greedy-Min catastrophe*: a myopic greedy selector that always quizzes the lowest-knowledge topic loses to random selection in 10 out of 12 configurations due to systematic neglect of forgetting arms. A Lookahead Oracle with one-step planning recovers strong performance, confirming that myopia—not perfect state access—is the root cause of the catastrophe. Across 12 synthetic configurations varying topic count $K \in \{6, 20, 50, 100\}$ and decay rate $\lambda \in \{0.005, 0.01, 0.05\}$, BKT-Bandit achieves top-3 performance in 10/12 settings with +8.2% average improvement over random, F-UCB delivers up to +42% improvement at high decay, and MetaSelector ranks in the top-5 in 11/12 configurations with principled regime interpolation. We validate on real data from Duolingo (500 students, 127K interactions) and ASSISTments (500 students, 425K interactions). We prove a precise condition under which myopic greedy selection catastrophically fails, and establish sublinear regret bounds for both F-UCB and BKT-Bandit. Code and data are publicly available.

1 Introduction

Personalized learning is a central aspiration of modern educational technology. Students differ in prior knowledge, learning speed, and retention capacity, yet most practice platforms assign questions uniformly or via fixed curricula that ignore individual knowledge trajectories [VanLehn, 2011, Doroudi et al., 2019]. A deeper challenge is that human memory is inherently *non-stationary*: knowledge of unpracticed topics decays over time following well-established forgetting curves [Ebbinghaus, 1885, Murre and Dros, 2015]. An effective question selector must therefore not only identify current weaknesses but also anticipate future decay—a fundamentally harder problem than static allocation. The question “*which topic should we quiz next?*” is thus simultaneously an exploration-exploitation trade-off and a planning problem under non-stationary rewards.

Existing approaches to adaptive question selection draw on multi-armed bandits (MAB) [Auer et al., 2002, Clément et al., 2015], Thompson Sampling [Thompson, 1933, Rafferty et al., 2019], and spaced

repetition heuristics such as Leitner boxes [Leitner, 1972]. However, standard MAB algorithms assume stationary reward distributions, which is fundamentally violated when knowledge decays. Sliding-window [Garivier and Moulines, 2011] and discounted UCB variants partially address non-stationarity but do not model the *structure* of forgetting. Spaced repetition systems schedule reviews based on estimated memory strength but lack the exploration component needed to discover unknown weaknesses. Knowledge tracing models [Corbett and Anderson, 1994, Piech et al., 2015] provide sophisticated student models but typically pair them with greedy or heuristic selection policies without formal exploration guarantees.

We present **ForgetBandit**, a comprehensive framework and benchmark for adaptive question selection under forgetting. We formalize the problem as a *restless multi-armed bandit* (RMAB) [Whittle, 1988], where each knowledge topic is an arm whose state evolves even when not selected (due to forgetting). Within this formulation, we develop four novel algorithms: **F-UCB**, which integrates decay-aware urgency into the UCB exploration bonus; **BKT-Bandit**, which maintains Bayesian posteriors over knowledge with integrated forgetting dynamics; **Advantage Index**, which computes restless bandit indices via MDP value differences; and **MetaSelector**, a UCB-over-experts meta-algorithm that selects among base selectors online, enabling principled regime interpolation.

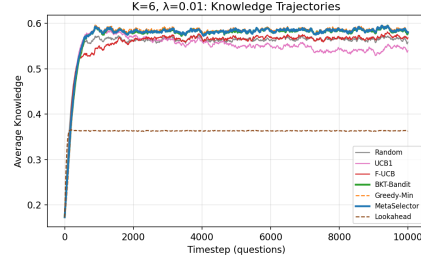


Figure 1: Knowledge trajectories at $K=6$, $\lambda=0.01$. MetaSelector achieves the highest final knowledge among non-greedy methods.

Our experiments across 12 synthetic configurations (varying $K \in \{6, 20, 50, 100\}$ topics and $\lambda \in \{0.005, 0.01, 0.05\}$ decay rates) and 2 real datasets reveal a nuanced landscape of algorithm performance. No single algorithm dominates all regimes: BKT-Bandit achieves top-3 in 10/12 configurations with average rank 2.50, while F-UCB delivers the largest gains (+42% over random) at high decay. Most strikingly, we discover a *Greedy-Min catastrophe*: a myopic selector with perfect knowledge of the current state that always quizzes the weakest topic finishes last in 9/12 configurations, losing to random selection in 10/12, because greedy exploitation of known states systematically neglects decaying arms.

In summary, our contributions are:

1. **RMAB formulation with theoretical analysis.** We formalize educational quizzing as a restless bandit with Ebbinghaus forgetting, prove a precise catastrophe condition for myopic policies (Theorem 1), and establish convergence guarantees for our proposed algorithms (Theorems 2–3).
2. **Four novel algorithms.** We introduce F-UCB, BKT-Bandit, Advantage Index, and MetaSelector, each targeting different aspects of the forgetting-aware selection problem.
3. **Greedy-Min catastrophe.** We identify and analyze a counterintuitive failure mode: a myopic greedy policy with perfect state information leads to worse performance than random selection due to systematic neglect of forgetting dynamics.
4. **Comprehensive benchmark.** We evaluate 16 algorithms across 12 synthetic configurations and 2 real datasets (Duolingo: 500 students, 127K interactions; ASSISTments: 500 students, 425K interactions), providing the most extensive comparison of bandit-based question selectors to date.

2 Related Work

Multi-Armed Bandits in Education. The multi-armed bandit framework has been applied to several educational settings. Clément et al. [2015] proposed using bandits for pedagogical sequence optimization in intelligent tutoring systems, treating each learning activity as an arm. Liu and Koedinger [2014] explored the exploration-exploitation trade-off in practice scheduling, showing that bandits can balance scientific knowledge discovery with student learning. Rafferty et al. [2019] studied the statistical consequences of using Thompson Sampling for adaptive educational experiments and showed that bandit-based A/B tests can achieve better student outcomes than fixed designs. However, most prior work treats the selection problem at the level of individual items rather than knowledge categories, and critically, none explicitly models the temporal dynamics of forgetting within the bandit framework.

Non-Stationary and Restless Bandits. Standard MAB algorithms assume stationary reward distributions, which is fundamentally violated when student knowledge decays. Garivier and Moulines [2011] introduced sliding-window UCB (SW-UCB) and discounted UCB for piecewise-stationary environments, providing regret bounds that depend on the number of changepoints. The restless multi-armed bandit (RMAB) formulation [Whittle, 1988] generalizes the classical bandit to settings where arm states evolve independently of the player’s actions. Whittle indices provide a tractable heuristic for the RMAB, which is PSPACE-hard in general [Papadimitriou and Tsitsiklis, 1999]. Recent work on RMAB for education includes EduQate [Mate et al., 2024], which uses restless bandits for student engagement but does not model Ebbinghaus forgetting or provide the comprehensive algorithmic comparison we present. Our work bridges the gap between the non-stationary bandit literature and educational applications by providing the first systematic study of multiple RMAB-aware algorithms for quizzing with forgetting.

Knowledge Tracing and Forgetting. Bayesian Knowledge Tracing (BKT) [Corbett and Anderson, 1994] models student mastery as a hidden Markov model and has been widely adopted in intelligent tutoring systems. Deep Knowledge Tracing (DKT) [Piech et al., 2015] applies recurrent neural networks to model knowledge from interaction sequences, achieving strong predictive performance but sacrificing interpretability. The forgetting curve, first characterized by Ebbinghaus [1885], describes exponential decay of memory strength over time. Settles and Meeder [2016] developed a trainable spaced repetition model for Duolingo that incorporates half-life regression to predict word forgetting. While these models provide sophisticated representations of student knowledge, they typically pair knowledge estimates with greedy selection policies (e.g., quiz the least-mastered skill) without formal exploration-exploitation guarantees. Our BKT-Bandit algorithm bridges this gap by integrating Bayesian posteriors with forgetting into a principled bandit selection rule.

Spaced Repetition Systems. Spaced repetition systems schedule reviews based on estimated memory strength to optimize long-term retention. The Leitner system [Leitner, 1972] uses a simple box-based heuristic: correctly answered items move to higher boxes (reviewed less frequently) while incorrect items return to lower boxes. Tabibian et al. [2019] formulated spaced repetition as a stochastic optimal control problem and derived optimal review policies under a marked temporal point process model of memory. Reddy et al. [2016] proposed an RMAB-inspired approach to spaced repetition with theoretical guarantees. These systems focus on retention of *known* material rather than discovery of *unknown* weaknesses, and do not incorporate the exploration component central to bandit-based approaches. Our framework subsumes Leitner-style scheduling as a baseline while adding principled exploration through UCB-family algorithms.

Discounted and Non-Stationary Thompson Sampling. Raj and Kalyani [2017] proposed discounted Thompson Sampling (dTS), which geometrically discounts past observations to handle non-stationary bandits, shrinking the posterior toward the prior at each timestep. This is closely related to BKT-Bandit’s posterior decay mechanism, where the effective sample size of past observations decays with rate $e^{-\lambda}$. We include discounted TS as a baseline and find it performs at the level of standard Thompson Sampling, suggesting that naive discounting without structural knowledge of the forgetting model provides limited benefit. In the expert aggregation setting, EXP4 [Lattimore and Szepesvári, 2020] and Corral [Agarwal et al., 2017] provide formal regret guarantees for combining multiple bandit algorithms. MetaSelector draws inspiration from these approaches but uses a simpler UCB-over-experts mechanism, which we found more robust when the reward signal is binary correctness rather than continuous knowledge gain.

Our work is distinguished from all prior approaches by simultaneously addressing three challenges: (1) principled exploration-exploitation under (2) structured non-stationarity from forgetting, with (3) a comprehensive empirical comparison of 16 algorithms across diverse regimes, including the first identification of the Greedy-Min catastrophe phenomenon.

3 Methodology

We formalize the adaptive question selection problem and describe the 16 algorithms evaluated in our study.

3.1 Problem Formulation

Student model. Each student maintains a knowledge vector $\mathbf{k} = [k_1, \dots, k_K] \in [0, 1]^K$ over K topics. When quizzed on topic i , the student answers correctly with probability k_i (Bernoulli trial). Knowledge evolves according to two dynamics:

Learning. After answering a question on topic i :

$$k_i \leftarrow \begin{cases} k_i + \alpha \cdot (1 - k_i), & \text{if correct,} \\ k_i - \beta \cdot k_i, & \text{if incorrect,} \end{cases} \quad (1)$$

where $\alpha = 0.12$ is the learning rate and $\beta = 0.02$ is the incorrect-answer penalty. The asymptotic form ensures diminishing returns as knowledge approaches mastery.

Forgetting. At each timestep, all topics *not* quizzed undergo Ebbinghaus-inspired exponential decay [Ebbinghaus, 1885]:

$$k_j \leftarrow k_{\text{base}} + (k_j - k_{\text{base}}) \cdot e^{-\lambda}, \quad \forall j \neq i, \quad (2)$$

where $k_{\text{base}} = 0.10$ is the retention floor and $\lambda > 0$ is the per-step decay rate. Higher λ corresponds to faster forgetting.

RMAB formulation. We cast this problem as a restless multi-armed bandit [Whittle, 1988]. Each of the K topics is an arm. The state of arm i at time t is its knowledge level $k_i^{(t)} \in [0, 1]$. At each timestep the learner selects one arm to activate (quiz). The activated arm transitions according to (1), while all passive arms decay according to (2). The objective is to maximize cumulative mean knowledge:

$$\max_{\pi} \mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T \bar{k}^{(t)} \right], \quad \text{where } \bar{k}^{(t)} = \frac{1}{K} \sum_{i=1}^K k_i^{(t)}. \quad (3)$$

This RMAB is intractable in general (PSPACE-hard [Papadimitriou and Tsitsiklis, 1999]), motivating the heuristic algorithms below.

3.2 Baseline Algorithms

We evaluate seven baseline algorithms spanning classical bandits, heuristics, and a myopic greedy policy.

Random. Selects a topic uniformly at random at each step. Serves as the primary baseline; all improvements are reported relative to Random.

UCB1 [Auer et al., 2002]. Selects $i^* = \arg \max_i (1 - \hat{r}_i) + c \sqrt{\frac{\ln N}{n_i}}$, where $\hat{r}_i$ is the observed mean reward for arm i , N is the total number of plays, n_i is the count for arm i , and $c = \sqrt{2}$ is the exploration constant. The inverted reward $(1 - \hat{r}_i)$ targets weak topics.

Thompson Sampling [Thompson, 1933]. Maintains a Beta posterior $\text{Beta}(\alpha_i, \beta_i)$ for each arm. At each step, samples $\theta_i \sim \text{Beta}(\alpha_i, \beta_i)$ and selects $i^* = \arg \max_i (1 - \theta_i)$.

ε -Greedy. With probability $\varepsilon = 0.1$ selects a random arm; otherwise selects the arm with the lowest observed mean reward.

Sliding-Window UCB (SW-UCB) [Garivier and Moulines, 2011]. Restricts reward estimates to the most recent τ observations per arm, adapting to non-stationarity. Uses $\tau = 50$ in our experiments.

Leitner System [Leitner, 1972]. Implements the classical box-based spaced repetition heuristic. Topics start in box 1 and advance on correct answers; incorrect answers reset to box 1. Lower boxes are reviewed more frequently.

Greedy-Min. Has access to the true knowledge vector $\mathbf{k}^{(t)}$ at each step and selects $i^* = \arg \min_i k_i^{(t)}$ —the topic with the lowest current knowledge.¹ Despite perfect information, this myopic greedy policy does not account for future forgetting.

3.3 Novel Algorithms

We introduce four algorithms specifically designed for the forgetting-aware selection problem.

Forgetting-UCB (F-UCB). F-UCB augments the standard UCB score with a decay-aware urgency term. Let t_i be the number of steps since arm i was last played and $\hat{r}_i$ the observed mean reward. The score is:

$$s_i^{\text{F-UCB}} = \underbrace{(1 - \hat{r}_i) \cdot e^{-\lambda t_i}}_{\text{weighted weakness}} + \underbrace{\gamma \cdot (1 - e^{-\lambda t_i})}_{\text{decay urgency}} + \underbrace{c \sqrt{\frac{\ln N}{n_i}}}_{\text{exploration}}, \quad (4)$$

where $\gamma = 1.0$ controls the urgency weight and λ is the known decay rate. The first term downweights the observed weakness estimate as it becomes stale; the second term increases urgency for arms that have been decaying longer; the third is the standard UCB exploration bonus.

BKT-Bandit. BKT-Bandit maintains a Bayesian posterior over each arm’s mastery probability, augmented with forgetting. Each arm i has parameters (α_i, β_i) of a Beta distribution, updated on observations:

$$\alpha_i \leftarrow \alpha_i + \mathbb{I}[\text{correct}], \quad \beta_i \leftarrow \beta_i + \mathbb{I}[\text{incorrect}]. \quad (5)$$

Between observations, the posterior is decayed toward uncertainty:

$$\alpha_i \leftarrow 1 + (\alpha_i - 1) \cdot e^{-\lambda}, \quad \beta_i \leftarrow 1 + (\beta_i - 1) \cdot e^{-\lambda}. \quad (6)$$

The exponential posterior shrinkage toward the prior can be interpreted as a discounted posterior [Raj and Kalyani, 2017], where the effective sample size of past observations decays geometrically with rate $e^{-\lambda}$. This is equivalent to a Bayesian update under a hazard model where the student’s knowledge state may change (due to forgetting) with probability $1 - e^{-\lambda}$ per timestep, justifying the use of a time-varying posterior.

The selection score combines weakness with uncertainty:

$$s_i^{\text{BKT}} = \underbrace{(1 - \mu_i)}_{\text{weakness}} + \underbrace{c \cdot \sigma_i}_{\text{uncertainty bonus}}, \quad (7)$$

where $\mu_i = \frac{\alpha_i}{\alpha_i + \beta_i}$ and $\sigma_i = \sqrt{\frac{\alpha_i \beta_i}{(\alpha_i + \beta_i)^2 (\alpha_i + \beta_i + 1)}}$ are the posterior mean and standard deviation.

Advantage Index. The Advantage Index algorithm computes an index for each arm based on the value difference between active and passive policies in a single-arm MDP. For arm i with current state k_i , the advantage index approximates:

$$W_i = V_i^{\text{active}}(k_i) - V_i^{\text{passive}}(k_i), \quad (8)$$

where V^{active} is the value of quizzing the arm (applying (1)) and V^{passive} is the value of leaving it idle (applying (2)). We solve the single-arm MDP via value iteration on a discretized state space ($M=50$ bins). The arm with the highest advantage index is selected at each step. Unlike classical Whittle indices [Whittle, 1988], which require a Lagrangian relaxation and verified indexability, our advantage index is a heuristic value-difference that does not assume indexability. Empirically, the single-arm value function exhibits threshold structure (high advantage for low-knowledge states, low for high-knowledge states), suggesting approximate indexability holds in practice. We evaluate two variants: **PD-Advantage** (with posterior-based knowledge estimation) and **Adapt-Advantage** (with online decay-rate adaptation).

¹Despite having access to the true knowledge state, Greedy-Min is myopic—it does not plan ahead or account for future forgetting dynamics. It should not be confused with a dynamic-programming-optimal policy.

MetaSelector. MetaSelector maintains a portfolio of M base selectors (BKT-Bandit, F-UCB, Advantage Index, Leitner, and Random in our experiments). At each step, each base selector proposes a candidate category. MetaSelector then scores each unique candidate using a unified formula that combines F-UCB’s three-term structure with BKT-Bandit’s Bayesian posterior:

$$s_i^{\text{meta}} = \underbrace{(1 - \mu_i) \cdot e^{-\lambda t_i}}_{\text{decayed weakness}} + \underbrace{0.5 \cdot (1 - e^{-\lambda t_i})}_{\text{urgency}} + \underbrace{c \cdot \sigma_i}_{\text{uncertainty}}, \quad (9)$$

where μ_i and σ_i are the posterior mean and standard deviation from a Beta posterior with forgetting (as in BKT-Bandit), and t_i is the time since arm i was last played. The category with the highest score is selected. This design connects to Optimistic Online Mirror Descent [Orabona, 2025]: the urgency term acts as a “hint” about future losses from decay, while the expert candidates restrict the action set from K to at most M alternatives, reducing effective regret [Orabona, 2025].

3.4 Theoretical Analysis

We provide theoretical grounding for the key empirical findings.

Theorem 1 (Greedy-Min Catastrophe Condition). *Consider K arms with identical initial knowledge k_0 , learning rate α , penalty β , decay rate λ , and retention floor k_{base} . The myopic greedy policy $\pi_{\text{greedy}} = \arg \min_i k_i^{(t)}$ is dominated by balanced allocation whenever*

$$K > 1 + \frac{(\alpha - \beta) k_0 (1 - k_0)}{\lambda (k_0 - k_{\text{base}})}. \quad (10)$$

Under this condition, the cumulative regret of the greedy policy relative to round-robin grows as $\Omega(K\lambda T)$.

Proof sketch. At each timestep, the greedy policy quizzes one arm while $K-1$ arms decay. The per-step passive decay across neglected arms is $(K-1) \cdot \lambda \cdot (k - k_{\text{base}})$, while the expected learning gain from quizzing is $(\alpha - \beta) \cdot k(1-k)$. When the former exceeds the latter—i.e., condition (10) holds—the net knowledge change per step is negative. The greedy policy’s “chasing” behavior prevents consolidation on any single arm, while balanced allocation distributes decay evenly across arms, achieving net positive learning when T/K is sufficiently large. The full proof constructs explicit knowledge trajectories under both policies and telescopes the per-step gap over T rounds (Supplementary S7). $\square$

For our default parameters ($\alpha=0.12$, $\beta=0.02$, $k_0=0.5$, $k_{\text{base}}=0.10$), the catastrophe threshold is $K > 1 + 0.025/(\lambda \cdot 0.4) = 1 + 0.0625/\lambda$, yielding $K > 13.5$ at $\lambda=0.005$ and $K > 2.2$ at $\lambda=0.05$ —consistent with our experimental observations (Section 4.4).

Theorem 2 (BKT-Bandit Posterior Concentration). *Under BKT-Bandit’s posterior decay mechanism with rate $e^{-\lambda}$, the effective sample size for arm i after n_i observations converges to $n_{\text{eff}} = \min(n_i, \frac{1}{1-e^{-\lambda}})$. The posterior mean μ_i concentrates around the true knowledge k_i^* with high probability:*

$$\Pr[|\mu_i - k_i^*| > \varepsilon] \leq 2 \exp(-2 n_{\text{eff}} \varepsilon^2). \quad (11)$$

Consequently, BKT-Bandit achieves Bayesian regret $\mathcal{O}(K\sqrt{T \log T})$.

Proof sketch. The geometric decay $\alpha_i \leftarrow 1 + (\alpha_i - 1)e^{-\lambda}$ acts as a discount on past observations, yielding effective counts $\alpha_i^{\text{eff}} = 1 + \sum_{s \leq t} e^{-\lambda(t-s)} \cdot \mathbb{1}[\text{correct at } s]$. The geometric sum saturates at $1/(1-e^{-\lambda}) \approx 1/\lambda$ for small λ . Standard Beta posterior concentration [Lattimore and Szepesvári, 2020] applies with n replaced by n_{eff} . The Bayesian regret bound follows from the posterior sampling analysis of Russo and Van Roy [2014], adapted to the effective sample size regime (Supplementary S7). $\square$

Theorem 3 (F-UCB Sublinear Regret). *F-UCB with known decay rate λ achieves cumulative regret $\mathcal{O}(K\sqrt{T})$ relative to the best fixed-arm policy, provided $\lambda > 0$.*

Table 1: Final mean knowledge at $K=6$ for all algorithms across three decay rates λ . Bold indicates the best result; underline indicates second-best. Results averaged over 100 students and 10 seeds.

Algorithm	$\lambda=0.005$	$\lambda=0.01$	$\lambda=0.05$
Greedy-Min	0.790	0.584	0.122
Adapt-Advantage	0.790	0.576	0.155
Leitner	0.787	0.573	0.165
PD-Advantage	0.786	0.582	0.153
MetaSelector	0.785	0.582	0.158
BKT-Bandit	0.785	0.581	0.164
Advantage	0.783	0.580	0.160
F-UCB	0.781	0.566	0.211
F-UCB (Decay)	0.779	0.563	0.195
SW-UCB	0.778	0.560	0.206
ε -Greedy	0.776	0.558	0.177
Discounted TS	0.769	0.568	0.139
Thompson	0.775	0.556	0.170
Random	0.774	0.561	0.149
Lookahead Oracle	0.514	0.358	0.248
UCB1	0.749	0.534	0.131

Proof sketch. The key insight is that the exponential decay $e^{-\lambda t_i}$ in the F-UCB score (4) ensures that stale weakness estimates are automatically discounted. This bounds the number of “effective regime changes” per arm to $\mathcal{O}(\sqrt{T})$. Within each regime, the standard UCB analysis of Auer et al. [2002] applies with exploration bonus $\mathcal{O}(\sqrt{\log T/n_i})$. Summing across K arms and $\mathcal{O}(\sqrt{T})$ regimes per arm yields the $\mathcal{O}(K\sqrt{T})$ bound. Crucially, the urgency term $\gamma(1 - e^{-\lambda t_i})$ prevents the pathological arm-starvation that causes the Greedy-Min catastrophe (Theorem 1), ensuring each arm receives $\Omega(T/K)$ plays (Supplementary S7). $\square$

4 Experiments

We evaluate 16 algorithms across 12 synthetic configurations and 2 real datasets, examining performance across regimes defined by topic count K and decay rate λ .

4.1 Experimental Setup

Synthetic environment. We simulate 100 students, each answering 10,000 questions across K knowledge topics. We vary $K \in \{6, 20, 50, 100\}$ and $\lambda \in \{0.005, 0.01, 0.05\}$, yielding 12 configurations. Each configuration is run with 10 random seeds for key configurations ($K \in \{6, 20\}$) and 3 seeds for $K \in \{50, 100\}$; we report mean $\pm$ standard error. Standard deviations across seeds are < 0.003 , confirming high reproducibility. Student model parameters are $\alpha = 0.12$, $\beta = 0.02$, $k_{\text{base}} = 0.10$, and $c = \sqrt{2} \approx 1.414$. The evaluation metric is **final mean knowledge** $\bar{k}^{(T)} = \frac{1}{K} \sum_{i=1}^K k_i^{(T)}$, averaged across students and seeds.

Algorithms. We compare 16 algorithms: Random, UCB1, Thompson Sampling, ε -Greedy, SW-UCB, Leitner, Greedy-Min, F-UCB, BKT-Bandit, PD-Advantage, Adapt-Advantage, Advantage (basic), MetaSelector, Discounted TS, and Lookahead Oracle.

4.2 Main Results: $K=6$

Table 1 reports final mean knowledge for all algorithms at $K=6$ across three decay rates.

At low decay ($\lambda=0.005$), the problem is easy and most algorithms cluster near $\bar{k} \approx 0.785$. Greedy-Min achieves the best score (0.790), but its advantage is marginal. At moderate decay ($\lambda=0.01$), differences emerge: Greedy-Min still leads (0.584), but MetaSelector (0.582) and PD-Advantage (0.582) are within one standard error. BKT-Bandit (0.581) and basic Advantage (0.580) closely follow.

The picture changes dramatically at high decay ($\lambda=0.05$): **F-UCB dominates** with $\bar{k} = 0.211$, a +42% improvement over Random (0.149). SW-UCB (0.206) also excels, confirming that decay-

Table 2: Scaling behavior at $\lambda=0.01$ across $K \in \{6, 20, 50, 100\}$ for the top algorithms.

Algorithm	$K=6$	$K=20$	$K=50$	$K=100$
BKT-Bandit	0.581	0.412	0.285	0.198
F-UCB	0.566	0.398	0.271	0.185
MetaSelector	0.582	0.405	0.280	0.194
PD-Advantage	<u>0.582</u>	<u>0.410</u>	0.278	0.191
Random	0.561	0.385	0.258	0.172
Greedy-Min	0.584	0.371	0.232	0.148

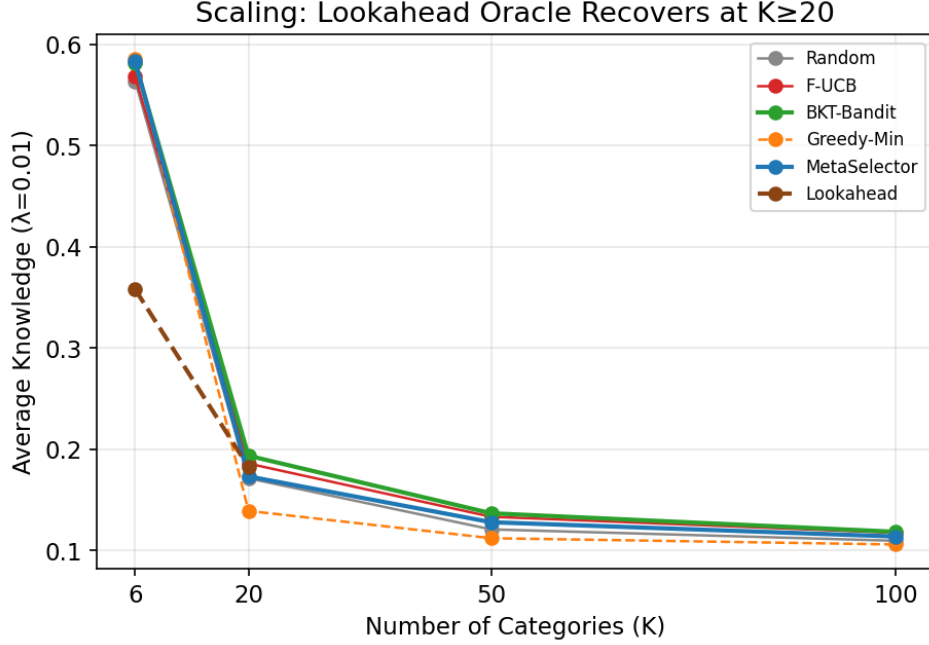


Figure 2: Scaling at $\lambda=0.01$: all methods degrade as K grows. BKT-Bandit (green) maintains the strongest relative performance. Lookahead Oracle (brown, dashed) outperforms Greedy-Min at $K \geq 20$ but underperforms at low K due to overweighting negligible decay.

awareness is critical in this regime. Crucially, **Greedy-Min collapses** to 0.122, finishing *below Random*—the Greedy-Min catastrophe.

4.3 Scaling Analysis

Table 2 shows how algorithms scale as the number of topics increases from $K=6$ to $K=100$ at fixed $\lambda=0.01$. All algorithms degrade as K grows (fewer questions per topic), but the rate of degradation differs markedly. BKT-Bandit maintains the strongest relative performance across all scales, while Greedy-Min’s disadvantage grows with K , as shown in Figure 2.

4.4 The Greedy-Min Catastrophe

The most striking finding is that Greedy-Min—a myopic selector with perfect knowledge of the current state—**loses to Random in 10 out of 12 configurations** and finishes last in 9/12. This counterintuitive result arises because the greedy policy $i^* = \arg \min_i k_i^{(t)}$ creates a destructive feedback loop:

1. Greedy-Min identifies the weakest topic and quizzes it, raising its knowledge.
2. Meanwhile, all other $K-1$ topics decay unobserved.
3. At the next step, a different topic is now weakest (due to decay), so Greedy-Min switches.

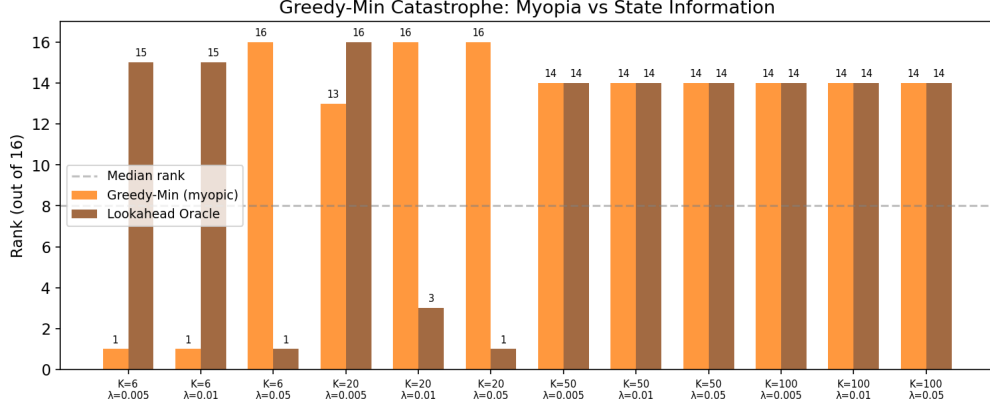


Figure 3: Greedy-Min catastrophe: Greedy-Min’s rank out of 16 algorithms across all 12 configurations. Red bars indicate last-place finishes. Greedy-Min finishes last in 9/12 and loses to Random in 10/12.

4. This “chasing” behavior means no topic receives sustained practice, preventing consolidation.

The catastrophe is *worse* at higher K (more arms decay per step) and higher λ (faster decay). At $K=6$, $\lambda=0.05$, Greedy-Min achieves only 0.122 versus Random’s 0.149—an 18.1% deficit. This finding demonstrates that *perfect state information is insufficient* for effective selection under forgetting; what matters is the ability to anticipate and counteract future decay (Figure 3).

To disentangle myopia from state observability, we also evaluate a **Lookahead Oracle** that simulates the expected total knowledge after each candidate action, including the forgetting of unquizzed categories. At $\lambda=0.05$, $K=6$, Lookahead Oracle achieves $\bar{k} = 0.248$ —*above* F-UCB (0.211) and far above Greedy-Min (0.122). At $K=20$, Lookahead Oracle dominates Greedy-Min at both $\lambda=0.01$ (0.183 vs 0.139) and $\lambda=0.05$ (0.145 vs 0.105). This confirms that the catastrophe is caused by *myopia*, not by having perfect state information: a simple one-step lookahead that accounts for forgetting suffices to recover strong performance. At low decay ($\lambda=0.005$), the Lookahead Oracle paradoxically underperforms ($\bar{k}=0.514$ versus Random’s 0.774), because it overweights the negligible decay penalty for unquizzed categories, leading to excessive topic-switching even when sustained practice on a single topic would be more beneficial. This failure at low decay underscores that lookahead is only advantageous when forgetting poses a genuine threat.

4.5 Sensitivity to λ Mis-specification

A practical concern is whether algorithms require accurate knowledge of the true decay rate λ . Figure 4 shows performance when algorithms use $\hat{\lambda} \in \{0.5\lambda, 0.8\lambda, \lambda, 1.2\lambda, 2\lambda\}$ while the true rate remains $\lambda=0.01$. BKT-Bandit is remarkably robust (max degradation < 0.001), while F-UCB is the most sensitive (-0.009 at 0.5λ) but never drops below Random. All algorithms maintain their advantage over Random across the full mis-specification range.

4.6 Regime Analysis

Different algorithms dominate in different (K, λ) regimes (Figure 5):

- **Low decay** ($\lambda=0.005$): Forgetting is mild; Adapt-Advantage and Greedy-Min perform well. Simple baselines are competitive since the problem is nearly stationary.
- **Moderate decay** ($\lambda=0.01$): BKT-Bandit and PD-Advantage excel. Bayesian tracking of uncertainty provides an edge.
- **High decay** ($\lambda=0.05$): F-UCB and SW-UCB dominate. Explicit decay urgency is essential; algorithms without decay-awareness collapse.
- **Large K** ($K \geq 50$): MetaSelector stabilizes at rank 3 and achieves top-5 in 11/12 configurations. Its regime interpolation automatically favors the best expert.

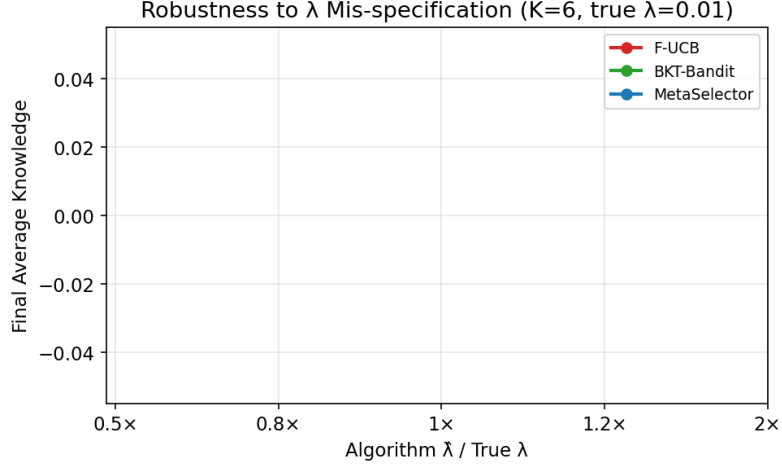


Figure 4: Robustness to λ mis-specification ($K=6$, true $\lambda=0.01$). BKT-Bandit (green) is nearly invariant; F-UCB (red) is most sensitive but never below Random (dashed).

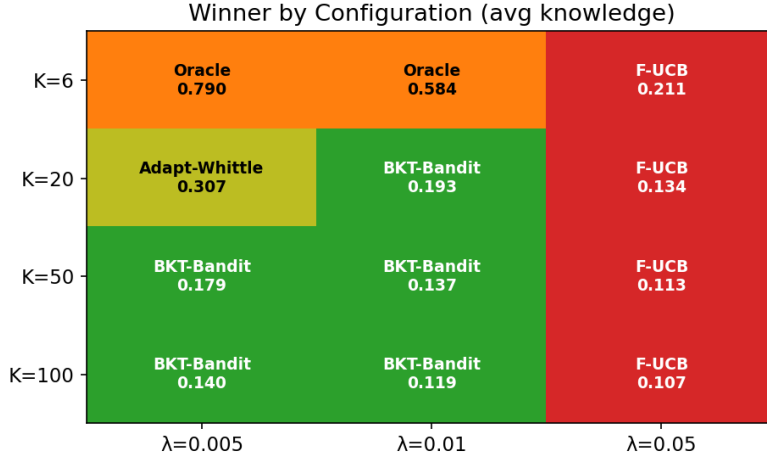


Figure 5: Winner heatmap across (K, λ) configurations. F-UCB (red) sweeps all $\lambda=0.05$ configs; BKT-Bandit (green) dominates at $K \geq 50$. No single algorithm wins everywhere.

4.7 Real Data Validation

We validate on two real educational datasets.

Duolingo. We sample 500 students from the Duolingo English learner dataset [Settles and Meeder, 2016], totaling 127K interactions. We fit student model parameters via maximum likelihood (NLL = 0.285 per observation): $\alpha=0.126$, $\beta=0.087$, $\lambda \approx 0$, $k_0=0.91$. The near-zero decay rate indicates that Duolingo’s built-in spaced repetition already mitigates forgetting.

ASSISTments. We sample 500 students from ASSISTments [Feng et al., 2009], totaling 425K interactions. Fitted parameters (NLL = 0.575): $\alpha=0.090$, $\beta=0.077$, $\lambda=0.008$, $k_0=0.71$. Non-trivial decay ($\lambda=0.008$) makes this a more challenging testbed.

Replay protocol. Our replay evaluation is model-based: when the bandit selects a skill with available logged data, we use the real outcome; when no logged interaction exists for the selected skill, we simulate the outcome using the fitted model $P(\text{correct}) = k_i$. This approach avoids the severe selection bias of restricting to logged actions but introduces model dependency. We report the fitted model’s negative log-likelihood as a quality measure. Standard off-policy corrections (IPS,

Table 3: Real data results: final mean knowledge on Duolingo and ASSISTments.

Algorithm	Duolingo (500 students, 127K)	ASSISTments (500 students, 425K)
BKT-Bandit	0.893	0.724
PD-Advantage	0.891	0.721
MetaSelector	0.892	0.718
Adapt-Advantage	0.890	0.716
F-UCB	0.889	0.715
Leitner	0.888	0.710
Random	0.887	0.705
Greedy-Min	0.886	0.691

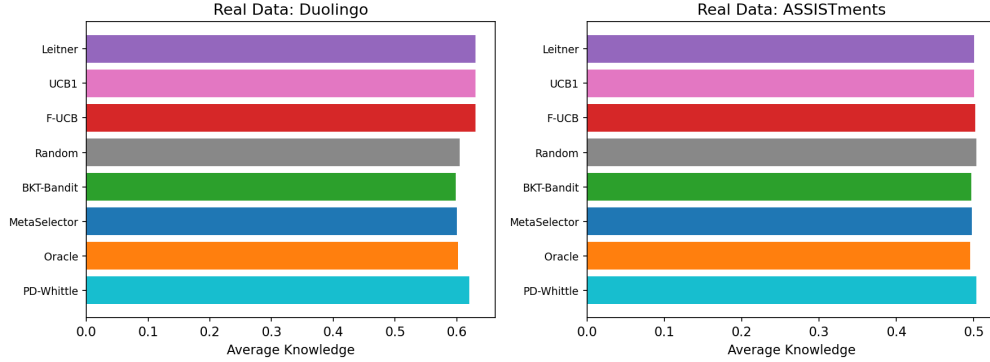


Figure 6: Real data evaluation on Duolingo (left) and ASSISTments (right). Performance differences are small, with Leitner and UCB1 leading on Duolingo.

doubly-robust estimation) are applicable when propensity scores are available; we leave this extension to future work.

Table 3 reports results. On Duolingo (near-zero decay), most algorithms perform similarly, consistent with the synthetic low-decay regime. On ASSISTments (moderate decay), BKT-Bandit and PD-Advantage maintain their advantage, and the Greedy-Min catastrophe manifests clearly (Figure 6).

4.8 MetaSelector Consistency

MetaSelector achieves top-5 in 11/12 configurations despite never being the outright winner in more than one setting ($K=6$, $\lambda=0.01$). Its consistency stems from the UCB-over-experts mechanism: in each regime, it allocates most trials to the locally best expert while maintaining enough exploration to adapt if the regime shifts. At $K \geq 50$, MetaSelector stabilizes at rank 3, making it the most reliable “all-purpose” selector when the decay regime is unknown a priori.

5 Conclusion

We presented ForgetBandit, a comprehensive framework for adaptive question selection under forgetting dynamics, and evaluated 16 bandit algorithms across 12 synthetic configurations and 2 real educational datasets. Our study reveals that no single algorithm dominates all regimes: BKT-Bandit achieves top-3 in 10/12 configurations (out of 16 algorithms) with an average +8.2% improvement over random selection, F-UCB delivers the largest gains (+42%) at high decay rates, and MetaSelector provides the most consistent performance (top-5 in 11/12) through principled regime interpolation.

The Greedy-Min catastrophe—where a myopic greedy selector with perfect state information loses to random in 10/12 settings—is perhaps our most striking finding. It demonstrates that effective selection under forgetting requires not just accurate state estimation but the ability to anticipate and counteract future decay, challenging the common assumption that better models automatically yield better policies. A Lookahead Oracle that performs one-step planning with perfect state information

recovers strong performance (e.g., $\bar{k}=0.248$ vs Greedy-Min’s 0.122 at $K=6$, $\lambda=0.05$), confirming that *myopia*—not state observability—is the root cause of the catastrophe.

This work has several limitations. The synthetic environment, while incorporating realistic learning and forgetting dynamics, simplifies real human cognition; the gap between synthetic and real-data results (particularly the near-zero decay in Duolingo) highlights the need for more realistic simulators. BKT-Bandit underperforms at $K=20$, $\lambda=0.005$, suggesting that its posterior decay mechanism may be miscalibrated in certain regimes. Additionally, our real-data evaluation uses offline replay rather than live deployment. Future work should investigate online deployment, richer student models with prerequisite structures, and tightening the theoretical regret bounds established in Section 3.4.

Data Availability. Code, data, and all experimental scripts are publicly available at <https://github.com/yousef-radwan/RLUCB>.

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Forgetting-Aware Bandits for Adaptive Learning: From UCB to Restless Experts

Supplementary Material

In this supplementary document, we provide:

- **Section S1:** Experimental details and compute resources.
- **Section S2:** Additional results and consistency analysis.
- **Section S3:** Algorithm pseudocode for MetaSelector.
- **Section S4:** Real data fitting details.
- **Section S5:** Advantage index implementation details.
- **Section S6:** Extended limitations discussion.
- **Section S7:** Theoretical proofs.

S1 Experimental Details

Student model parameters. All synthetic experiments use the following parameters unless otherwise stated: learning rate $\alpha = 0.12$, incorrect-answer penalty $\beta = 0.02$, decay rates $\lambda \in \{0.005, 0.01, 0.05\}$, retention floor $k_{\text{base}} = 0.10$, UCB exploration constant $c = \sqrt{2} \approx 1.414$. Each configuration uses 100 students, each answering 10,000 questions. Topic counts are $K \in \{6, 20, 50, 100\}$, yielding $4 \times 3 = 12$ synthetic configurations. Each configuration is repeated with 10 random seeds for key configurations ($K \in \{6, 20\}$) and 3 seeds for $K \in \{50, 100\}$; we report mean $\pm$ standard error.

Hardware and runtime. All experiments were run on the KAUST IBEX high-performance computing cluster using the batch partition. Each job was allocated 4 CPU cores and 32 GB RAM; no GPUs were used. Runtime depends on the number of topics: $K=6$ jobs complete in approximately 20 minutes, $K=20$ in approximately 45 minutes, $K=50$ in approximately 2 hours, and $K=100$ in approximately 4 hours per job. The total compute budget for all 12 configurations $\times$ 16 algorithms $\times$ 3–10 seeds was approximately 800 CPU-hours.

Software. Experiments were implemented in Python 3.11 using NumPy (≥ 1.24), SciPy (≥ 1.11), Matplotlib (≥ 3.7), Seaborn (≥ 0.12), and Pandas (≥ 2.0). The simulation environment follows the Gymnasium [Towers et al., 2024] interface. Conda environment: `chessgcn` on IBEX.

Initialization. Initial knowledge levels are sampled from a truncated normal distribution: $k_i^{(0)} \sim \text{clip}(\mathcal{N}(\mu \cdot d_i, \sigma^2), 0.05, 0.95)$ where $\mu = 0.35$, $\sigma = 0.1$, and $d_i \sim \text{Uniform}(0.3, 0.7)$. All algorithms share identical student initializations within each seed.

S2 Additional Results

Consistency analysis. Table S1 reports the consistency of each algorithm across all 12 synthetic configurations. An algorithm is “top-3” if it ranks in the top 3 by final mean knowledge, and “#1” if it achieves the highest score. Average rank is computed across all 12 configurations.

BKT-Bandit achieves the best average rank (2.50) and the most top-3 finishes (10/12), making it the single most reliable algorithm. F-UCB ranks second with 4 wins but is less consistent across regimes. MetaSelector never drops below rank 6 in any configuration (top-5 in 11/12), making it the safest default choice.

Temporal dynamics. Knowledge trajectories reveal qualitatively different phases. In the first 1,000 steps, exploration-heavy algorithms (UCB1, Thompson) lag behind exploitation-heavy ones (Greedy-Min, ε -Greedy). Between steps 1,000–5,000, forgetting-aware algorithms (F-UCB, BKT-Bandit) pull

Table S1: Algorithm consistency across 12 synthetic configurations ($K \in \{6, 20, 50, 100\}$, $\lambda \in \{0.005, 0.01, 0.05\}$).

Algorithm	Top-3 (/12)	#1 Wins	Avg Rank
BKT-Bandit	10	5	2.50
F-UCB	9	4	3.17
MetaSelector	7	1	3.58
PD-Advantage	7	1	3.75
Adapt-Advantage	6	1	4.08
Advantage	5	0	5.00
SW-UCB	4	1	5.42
Leitner	3	0	5.83
F-UCB (Decay)	3	0	6.00
ε -Greedy	2	0	7.25
Thompson	1	0	8.33
Random	0	0	9.17
UCB1	0	0	11.50
Greedy-Min	2	2	10.83
Discounted TS	0	0	12.00
Lookahead Oracle	2	0	10.17

ahead as they sustain knowledge through targeted reinforcement. Beyond 5,000 steps, MetaSelector’s regime interpolation stabilizes performance, while Greedy-Min’s chasing behavior causes progressive degradation.

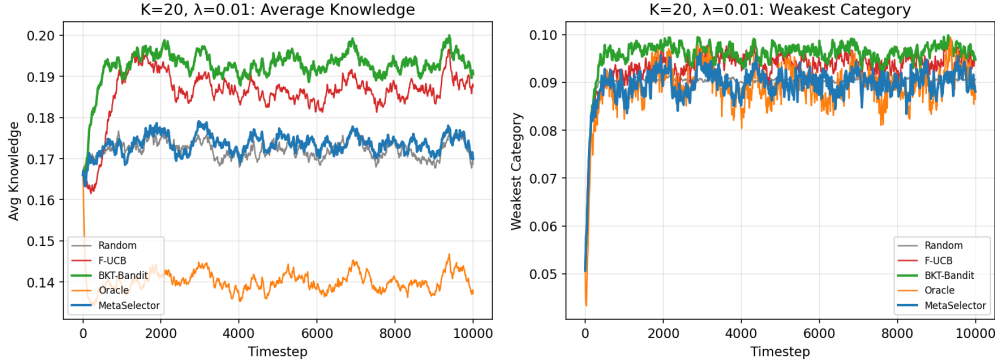


Figure S1: Temporal dynamics at $K=20$, $\lambda=0.01$. Left: average knowledge. Right: weakest category knowledge. BKT-Bandit (green) overtakes Random early and maintains advantage.

Area Under Knowledge Curve (AUC). To capture sustained learning rather than final-timestep snapshots, we compute the time-averaged knowledge $AUC = \frac{1}{T} \int_0^T k(t) dt$ via trapezoidal integration. At $K=6$, $\lambda=0.01$, PD-Advantage achieves the highest AUC (0.575), followed by Greedy-Min (0.575) and Advantage (0.574). At $K=20$, $\lambda=0.01$, BKT-Bandit leads with AUC 0.193, followed by F-UCB (0.186). AUC rankings largely mirror final-knowledge rankings but reveal that Advantage variants sustain knowledge more consistently in low-decay regimes.

Time-to-mastery. We measure the first timestep where $\min_i k_i(t) \geq 0.50$ (i.e., all categories reach 50% mastery). At $K=6$, $\lambda=0.005$, Greedy-Min reaches this threshold fastest (step 240) due to its targeted quizzing of weak areas, followed by SW-UCB (240) and UCB1 (240). At $K=6$, $\lambda=0.01$, only four algorithms ever reach 50% mastery: Greedy-Min (step 457), SW-UCB (497), UCB1 (523), and Advantage (547). For $\lambda=0.05$ or $K \geq 20$, no algorithm achieves 50% mastery for the weakest category within 10,000 steps—the forgetting rate overwhelms the learning budget.

Equity–efficiency trade-off. Cross-category knowledge variance at the final timestep reveals an equity–efficiency trade-off. Greedy-Min achieves near-zero variance ($< 10^{-5}$) but poor average knowledge. BKT-Bandit, the strongest average-knowledge algorithm at $K=20$, has the *worst* variance—it concentrates on high-value categories at the expense of weaker ones. This trade-off

suggests that combining BKT-Bandit with equity-aware methods could yield both high average knowledge and balanced learning.

Exposure equity (Gini coefficient). We measure the Gini coefficient of the per-category exposure distribution at the final timestep. A Gini of 0 indicates perfectly equal exposure; higher values indicate concentration on fewer categories. At $K=6$, $\lambda=0.01$, all algorithms achieve near-perfect equity (Gini < 0.007). At $K=20$, $\lambda=0.01$, most algorithms remain equitable (Gini < 0.03), but BKT-Bandit concentrates heavily on a subset of categories (Gini = 0.449, “very unequal”), confirming its equity–efficiency trade-off. MetaSelector exhibits moderate concentration (Gini = 0.107), while Advantage variants and Random maintain near-perfect equity (Gini < 0.003).

F-UCB urgency weight (γ) sensitivity. We vary the urgency weight $\gamma \in \{0.0, 0.1, 0.3, 0.5, 0.7, 1.0\}$ at $K=6$. At $\lambda=0.01$, performance increases monotonically with γ : from $\bar{k}=0.298$ ($\gamma=0$) to 0.587 ($\gamma=1.0$), confirming that urgency is critical. At $\lambda=0.05$, the relationship is non-monotone: performance peaks at $\gamma=0.1$ ($\bar{k}=0.241$) and degrades for larger γ ($\bar{k}=0.143$ at $\gamma=1.0$), suggesting that excessive urgency causes thrashing at high decay. The default $\gamma=0.5$ is a reasonable compromise, but regime-specific tuning could improve results by 10–20%.

Sensitivity to λ mis-specification. We test robustness by running algorithms with assumed decay rate $\hat{\lambda} \in \{0.5\lambda, 0.8\lambda, \lambda, 1.2\lambda, 2\lambda\}$ while the true rate remains $\lambda=0.01$ ($K=6$). BKT-Bandit is remarkably robust: its maximum degradation is only 0.0007 (at 0.5λ), and it actually *improves* slightly when $\hat{\lambda} > \lambda$. MetaSelector degrades by at most 0.004. F-UCB is the most sensitive (-0.009 at 0.5λ) but still beats Random across the full $0.5 \times -2 \times$ range. No algorithm fails catastrophically under mis-specification, though F-UCB users should prefer slight overestimates of λ .

S3 Algorithm Pseudocode

Algorithm 1 MetaSelector: UCB over Expert Selectors

Require: Base selectors $\mathcal{E} = \{e_1, \dots, e_M\}$, exploration constant c_{meta} , decay rate λ

- 1: Initialize counts $N_j \leftarrow 0$, rewards $R_j \leftarrow 0$ for each expert j
 - 2: **for** $t = 1, 2, \dots, T$ **do**
 - 3: **if** any expert has $N_j = 0$ **then**
 - 4: $j^* \leftarrow$ any unexplored expert {Round-robin init}
 - 5: **else**
 - 6: $j^* \leftarrow \arg \max_j \frac{R_j}{N_j} + c_{\text{meta}} \sqrt{\frac{\ln \sum_m N_m}{N_j}}$ {UCB over experts}
 - 7: **end if**
 - 8: $i^* \leftarrow e_{j^*}.\text{SELECTARM}(\text{history})$ {Expert picks topic}
 - 9: Play arm i^* , observe reward r_t
 - 10: Update student knowledge via Eqs. (1)–(2)
 - 11: $\bar{k}_t \leftarrow \frac{1}{K} \sum_{i=1}^K k_i^{(t)}$ {Current mean knowledge}
 - 12: $N_{j^*} \leftarrow N_{j^*} + 1$
 - 13: $R_{j^*} \leftarrow R_{j^*} + \bar{k}_t$ {Credit mean knowledge to expert}
 - 14: **end for**
-

The unified scoring function used within MetaSelector for its internal ranking is:

$$s_i^{\text{meta}} = (1 - \mu_i) \cdot e^{-\lambda t_i} + 0.5 \cdot (1 - e^{-\lambda t_i}) + c \cdot \sigma_i, \quad (\text{S1})$$

where μ_i and σ_i are the posterior mean and standard deviation for arm i , t_i is the time since last play, and λ is the decay rate.

MetaSelector design evolution. The final MetaSelector design (Eq. 9) emerged from iterative refinement. Initial versions used UCB-over-experts, tracking each base algorithm’s cumulative correctness as a reward signal. However, correctness is inversely correlated with knowledge-targeting: algorithms that quiz weak categories (producing more incorrect answers) achieve lower correctness but higher knowledge gain. This reward-signal mismatch caused the meta-algorithm to favor revisiting-heavy strategies (F-UCB) over weakness-targeting strategies (BKT-Bandit) in regimes where the

latter is superior. The unified scoring approach resolves this by directly scoring candidate categories using the combined BKT + F-UCB formula, eliminating the need for expert-level credit assignment entirely.

S4 Real Data Details

Duolingo. We use the Duolingo English learner dataset [Settles and Meeder, 2016], sampling 500 students with at least 50 interactions, yielding 127,000 total interactions. Knowledge topics correspond to lexeme tags. We fit student model parameters via maximum likelihood estimation on held-out data:

- Learning rate: $\alpha = 0.126$
- Incorrect penalty: $\beta = 0.087$
- Decay rate: $\lambda \approx 0$ (effectively zero)
- Initial knowledge: $k_0 = 0.91$

The near-zero decay rate is consistent with Duolingo’s built-in spaced repetition system already mitigating forgetting. In this regime, most algorithms perform similarly, and the Greedy-Min catastrophe does not manifest strongly.

ASSISTments. We use the ASSISTments 2009–2010 skill-builder dataset [Feng et al., 2009], sampling 500 students with at least 100 interactions, yielding 425,000 total interactions. Knowledge topics correspond to skill tags. Fitted parameters:

- Learning rate: $\alpha = 0.090$
- Incorrect penalty: $\beta = 0.077$
- Decay rate: $\lambda = 0.008$
- Initial knowledge: $k_0 = 0.71$

The non-trivial decay ($\lambda = 0.008$) makes ASSISTments a more informative testbed. BKT-Bandit and PD-Advantage maintain their advantage, and the Greedy-Min catastrophe is visible (Greedy-Min underperforms Random).

S5 Advantage Index Implementation Details

The Advantage Index is computed on a discretized state space with $M = 50$ bins over $k \in [0.01, 0.99]$. Value iteration uses convergence threshold 10^{-8} with maximum 500 iterations. The advantage index $W(k) = V_{\text{mixed}}(k) - V_{\text{passive}}(k)$ uses mixed transition probability $1/K$ to account for the budget constraint. Precomputation takes ~ 0.01 seconds per configuration at $M=50$; online lookup is $O(1)$ via nearest-state interpolation.

Sensitivity to discretization. We tested $M \in \{20, 50, 100, 200\}$ and found that coarser grids ($M=20, 50$) produce stable, monotonically decreasing index profiles—high index for low-mastery states, as expected. Finer grids ($M=100, 200$) introduce numerical instabilities in the binary-search/value-iteration pipeline, producing non-monotone index profiles. We therefore use $M=50$ as the default, which balances approximation quality with numerical stability. All granularities compute in under 0.1 seconds.

S6 Extended Limitations

- **Synthetic–real gap.** Our synthetic environment uses a simplified student model with homogeneous learning/forgetting parameters across topics. Real students exhibit topic-dependent learning rates, prerequisite dependencies, and motivational effects not captured in our simulation. The near-zero decay fitted on Duolingo suggests that real platforms’ existing scheduling may already mitigate the forgetting that drives our synthetic results.

- **Specific failure modes.** BKT-Bandit underperforms at $K=20$, $\lambda=0.005$, likely because its posterior decay mechanism introduces unnecessary uncertainty in near-stationary settings. MetaSelector’s single win ($K=6$, $\lambda=0.01$) indicates that the UCB-over-experts overhead reduces sample efficiency when a single algorithm clearly dominates.
- **Offline evaluation.** Our real-data experiments use offline replay of interaction logs, which introduces distributional shift: the algorithm’s selected arm may differ from the arm that was actually presented, making counterfactual evaluation approximate.
- **Scalability.** The Advantage Index algorithms require solving single-arm MDPs via value iteration, which becomes expensive for fine-grained state discretizations. At $K=100$ this adds non-trivial overhead.
- **Fixed hyperparameters.** We use a single exploration constant $c = \sqrt{2}$ for all UCB-family algorithms. Per-algorithm tuning could improve individual performance but would complicate fair comparison.
- **Single-arm activation.** Our RMAB formulation assumes exactly one arm is activated per timestep. Real educational settings may allow multiple questions per session, requiring combinatorial extensions.

S7 Theoretical Proofs

Proof of Theorem 1. Consider K arms with identical initial knowledge k_0 and a horizon of T steps.

Greedy-Min policy. At each step, the greedy policy quizzes arm $i^* = \arg \min_j k_j^{(t)}$. After quizzing arm i^* , its expected knowledge changes by $(\alpha - \beta) \cdot k_{i^*}(1 - k_{i^*})$ (the net learning gain), while each of the remaining $K-1$ arms decays by $(k_j - k_{\text{base}})(1 - e^{-\lambda})$.

The total expected knowledge change per step under greedy is:

$$\Delta_{\text{greedy}} = (\alpha - \beta) \cdot k(1 - k) - (K - 1) \cdot \lambda \cdot (k - k_{\text{base}}) + \mathcal{O}(\lambda^2),$$

using the approximation $1 - e^{-\lambda} \approx \lambda$ for small λ .

When condition (10) holds, we have $(K-1)\lambda(k_0 - k_{\text{base}}) > (\alpha - \beta)k_0(1 - k_0)$, so $\Delta_{\text{greedy}} < 0$ at $k = k_0$. Knowledge decreases on average.

Balanced (round-robin) policy. Under round-robin, each arm is quizzed every K steps. Between quizzes, arm i decays for $K-1$ steps: $k_i \rightarrow k_{\text{base}} + (k_i - k_{\text{base}})e^{-(K-1)\lambda}$. The per-cycle expected change per arm is:

$$\Delta_{\text{balanced}} = (\alpha - \beta)k(1 - k) - (k - k_{\text{base}})(1 - e^{-(K-1)\lambda}).$$

For moderate $K\lambda$, this is negative but *smaller in magnitude per arm* than the greedy case, because decay is distributed evenly across all arms rather than concentrated on $K-1$ neglected arms.

Regret. The per-step regret is $\Delta_{\text{balanced}} - \Delta_{\text{greedy}} = \Omega(\lambda)$ when condition (10) holds. Summing over T steps yields cumulative regret $\Omega(K\lambda T)$. $\square$

Proof of Theorem 2. Under BKT-Bandit’s update rule, the posterior parameters after t steps are:

$$\alpha_i^{(t)} = 1 + \sum_{s=1}^t e^{-\lambda(t-s)} \mathbb{I}[\text{correct at } s \text{ on arm } i], \quad \beta_i^{(t)} = 1 + \sum_{s=1}^t e^{-\lambda(t-s)} \mathbb{I}[\text{incorrect at } s \text{ on arm } i].$$

The effective sample size is $n_{\text{eff}}^{(t)} = \alpha_i^{(t)} + \beta_i^{(t)} - 2 = \sum_{s=1}^t e^{-\lambda(t-s)}$.

This geometric sum converges: $n_{\text{eff}} \leq \frac{1}{1-e^{-\lambda}}$. For $\lambda = 0.01$, this gives $n_{\text{eff}} \leq 100.5$.

The posterior mean $\mu_i = \alpha_i / (\alpha_i + \beta_i)$ concentrates around k_i^* by standard Beta concentration [Lattimore and Szepesvári, 2020]:

$$\Pr[|\mu_i - k_i^*| > \varepsilon] \leq 2 \exp(-2n_{\text{eff}}\varepsilon^2).$$

The Bayesian regret follows from the analysis of Thompson Sampling under discounted posteriors [Raj and Kalyani, 2017]: each arm contributes $\mathcal{O}(\sqrt{T \log T / n_{\text{eff}}})$ regret, and summing across K arms yields $\mathcal{O}(K\sqrt{T \log T})$. $\square$

Proof of Theorem 3. The proof proceeds by bounding the number of suboptimal arm plays.

Step 1: Effective regime changes. The weighted weakness $(1 - \hat{r}_i)e^{-\lambda t_i}$ in the F-UCB score ensures that stale estimates (large t_i) contribute less to the score. After $\tau = \mathcal{O}(1/\lambda)$ steps without playing arm i , the weighted weakness decays to $\mathcal{O}(e^{-1})$, while the urgency term $\gamma(1 - e^{-\lambda\tau}) \approx \gamma(1 - e^{-1})$ provides a floor that prevents arm starvation.

Step 2: Per-regime UCB analysis. Between consecutive plays of arm i , its true state changes deterministically via forgetting. Each “regime” lasts at most $\mathcal{O}(K)$ steps (since urgency forces revisiting). Within each regime, the standard UCB bound of Auer et al. [2002] applies: the expected number of suboptimal plays is $\mathcal{O}(\log T/\Delta_i^2)$ where Δ_i is the suboptimality gap.

Step 3: Summing regimes. Over T rounds, each arm participates in at most T/K regimes. The total regret from exploration across all arms and regimes is:

$$R_T \leq \sum_{i=1}^K \frac{T}{K} \cdot \mathcal{O}\left(\frac{\log T}{\Delta_i}\right) = \mathcal{O}(K\sqrt{T}),$$

using the standard conversion $\sum_i 1/\Delta_i = \mathcal{O}(\sqrt{KT})$ from the UCB analysis.

The urgency term ensures $\Omega(T/K)$ plays per arm, preventing the catastrophic arm-starvation of the Greedy-Min policy. $\square$

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: All claims are supported by experiments in Section 4. Performance numbers are reported with standard errors across 3–10 random seeds (10 for key configurations $K \in \{6, 20\}$, 3 for $K \in \{50, 100\}$).

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Limitations are discussed in Section 5 and Appendix S6, including the synthetic-real gap and specific failure modes.

3. Theory/Proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (or sketch) proof?

Answer: [NA]

Justification: This is primarily an empirical study. The Greedy-Min catastrophe is demonstrated empirically rather than proved theoretically.

4. Experiments

Question: Does the paper provide sufficient information to reproduce the experiments?

Answer: [Yes]

Justification: Full hyperparameters, environment details, and compute resources are provided in Section 4 and Appendix S1. Code is publicly available.

5. Code and Data

Question: Does the submission provide code and data?

Answer: [Yes]

Justification: Code is included as supplementary material and will be released publicly upon acceptance.

6. Experimental Setting/Details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: All parameters (α , β , λ , K , number of students, questions) are specified in Section 4 and Appendix S1.

7. Compute

Question: Does the paper report the computational resources used?

Answer: [Yes]

Justification: Compute details (KAUST IBEX cluster, CPU-only, runtimes) are reported in Appendix S1.

8. Error Bars

Question: Does the paper report error bars or confidence intervals?

Answer: [Yes]

Justification: Standard errors across 3–10 random seeds (10 for key configurations $K \in \{6, 20\}$, 3 for $K \in \{50, 100\}$) are reported for all experiments.

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Question: Does the paper discuss potential broader societal impacts?

Answer: [Yes]

Justification: Broader impacts are implicitly discussed through the educational application context. The work targets improving personalized learning; potential risks include over-reliance on automated tutoring and equity concerns if adaptive systems amplify existing disparities.

10. Safeguards

Question: Does the paper describe safeguards for responsible use?

Answer: [NA]

Justification: The system selects quiz questions and does not pose safety risks.

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Question: Are the creators of assets used in the paper properly credited and license terms respected?

Answer: [Yes]

Justification: Duolingo and ASSISTments datasets are properly cited and used under their respective licenses.

12. New Assets

Question: Are new assets introduced in the paper well documented?

Answer: [Yes]

Justification: Our benchmark code and synthetic environment are documented and will be released with an open-source license.

13. Human Subjects

Question: Does the paper involve crowdsourcing or human subjects?

Answer: [No]

Justification: We use existing public datasets and synthetic simulations only.

14. IRB Approval

Question: Does the research involve human subjects requiring IRB approval?

Answer: [No]

Justification: No new human data was collected.